Test ID	Target Function	Test Input Vector	Expected Output	Actual Output (Legacy Code)	Pass / Fail	Technical Remarks
TC01	add_item()	Add 10, 20, 30 (normal)	Items added; count = 3	Printed 'Added item' for each value	Pass	Normal add within initial capacity of 5 works.
TC02	add_item()	Add 6 items (triggers resize)	Capacity doubles to 10; all 6 items kept	All 6 items added and printed; old array not freed (seen in code, line 20)	Fail	Resize works but leaks the old buffer. Needs a memory tool to prove.
TC03	printAll()	Print list with 6 items	Shows 10 20 30 40 50 60 only	Observed: 10 20 30 40 50 60 1784086864	Fail	Off-by-one: i <= currentCount (line 56) reads past the data.
TC04	findItem()	Search for 30	Returns index 2	Observed: Found 30 at index: 2	Pass	Works despite the pointless inner loop.
TC05	findItem()	Search for 999 (missing)	Returns -1	Observed: Found 999 at index: -1	Pass	Returned -1 as expected. Non-existent element lookup handles return safely.
TC06	remove_item_at()	Delete valid index 2	30 removed; list = 10 20 40 50 60	Observed: 'Item removed from index 2'; printed 10 20 40 50 60 60 (last 60 is the over-read)	Pass	Removal itself works; the extra 60 comes from the printAll bug (TC03).
TC07	remove_item_at()	Delete negative index -5	Invalid index!' and no change	Observed: Printed 'Invalid index!'	Pass	Negative index check triggers successfully; state remains unchanged.
TC08	remove_item_at()	Delete out-of-bounds index 99	Invalid index!' and no change	Observed: 'Item removed from index 99'; count dropped by 1 anyway	Fail	No upper bound check; count is silently corrupted. It did not crash.
TC09	processMatrix()	Transpose 3x3 matrix	1 4 7 / 2 5 8 / 3 6 9	Observed: 1 4 7 / 2 5 8 / 3 6 9	Pass	Correct, but sizes are hardcoded and r, c are unused.
TC10	Memory cleanup	Program exit	Zero leaks reported	Memory leak: heap memory remains unallocated/unreleased on program exit	Fail	Lacks cleanup delete[] call on termination and leaks old buffer on resize.
TC11	remove_item_at()	Delete index 0 (head boundary)	First item removed; rest shift left	Observed: 'Item removed from index 0'; printed 20 30 0	Pass	First item removed and remaining elements shifted, but contains redundant nested loops.
TC12	printAll()	Print an empty list (nothing added)	Clean message, no crash	crashed (Segmentation fault)	Fail	Attempting to read uninitialized NULL dataArray pointer causes immediate program crash.